Northern Illinois University

# Adaptive Underwater Communication

*Project Report*

**Advisors**

Sachit Butail

Benedito Fonseca

# Table of Contents


1 Introduction ..... 1
1.0.1 Literature Review ..... 1

2 Multi-Robot Sampling ..... 5
2.1 Introduction ..... 5
2.2 Background ..... 5
2.2.1 Modulation Techniques in Underwater Acoustic Communication ..... 6
2.2.2 Binary Phase Shift Keying (BPSK) ..... 6
2.2.3 Quadrature Phase Shift Keying (QPSK) ..... 6
2.2.4 4-Quadrature Amplitude Modulation (4-QAM) ..... 6
2.2.5 Comparison and Applications ..... 6

3 Impact of obstacles on line of sight (LOS) in Under- water Communications ..... 7
3.1 Modelling Attenuation with Obstacle ..... 8
3.1.1 Diffraction ..... 8
3.1.2 LOS Transmission ..... 9
3.1.3 Reflections ..... 9
3.1.4 Total Attenuation ..... 9

4 Simulation ..... 11
4.0.1 Effect of Removing Front and Back Terms in Attenuation Model 11
4.0.2 Adaptive sampling: ..... 12
4.0.3 Model Predictive Control ..... 13
4.1 Adaptive Control for High SNR Communication ..... 13
4.1.1 Problem Statement: ..... 13

**4.2** Vehicle Dynamics: ... 14
**4.3** Optimal Control for Multi-AUV System ... 15
**4.3.1** Gradient Descent ... 15
**4.3.2** Model Predictive Control ... 16

**5 Hardware** ... **19**
**5.1** Power consumption analysis and runtime limitation ... 19
**5.1.1** Component selection and specifications ... 19
**5.1.2** System Architecture ... 19
**5.2** Implementation challenges and solutions ... 20

**6 Timeline** ... **23**
**6.0.1** Fall 2024 ... 23
**6.0.2** Spring 2025 ... 24
**6.0.3** Summer 2025 ... 24
**6.0.4** **Updated project summary** ... 25

**Bibliography** ... **27**

# CHAPTER 1
# Introduction

Underwater sampling is crucial for understanding and preserving marine environments, providing valuable data on water quality, biodiversity, and the impacts of climate change. This information is essential for environmental monitoring, scientific research, resource management, and climate studies. Traditionally, underwater sampling has been conducted using stationary sensors or human-operated vehicles, which are limited by coverage, accessibility, cost, and safety concerns. The deployment of multiple autonomous underwater robots addresses these limitations by increasing coverage, offering flexibility and adaptability, and operating more efficiently.

Among these robots, underwater gliders stand out due to their energy efficiency, ability to undertake extended missions, and capability to dive to great depths. These features enable data collection over vast areas and extended periods. Integrating multiple autonomous robots, especially underwater gliders, supported by robust communication and cooperative strategies, significantly enhances the efficiency, coverage, and reliability of underwater sampling, advancing our understanding and preservation of marine environments.

Effective communication between these robots is vital for coordinated operations, facilitating real-time data sharing, task coordination, and adaptive navigation. Cooperative strategies in communication, navigation, and path planning further enhance the effectiveness of underwater sampling. These strategies allow robots to exchange environmental information, maintain accurate positioning, and strategically divide sampling areas, ensuring comprehensive coverage and efficient resource use.

One of the most effective communication strategies for underwater gliders, proven to handle the vast and varied underwater conditions, is acoustic communication. The goal of this research is to: 1) Design a control strategy that can optimally sample underwater regions while maintaining robust communication in the presence of obstacles and clutter 2) Validate this strategy on a testbed consisting of both real and simulated underwater robots.

### 1.0.1 Literature Review

Historically, oceanographic research has been conducted through ship-based sampling methods, with one of the earliest global expeditions being Sir Wyville Thomson's HMS Challenger voyage from 1872 to 1876. Ship-based measurements and moorings were the primary means for studying temporal and spatial ocean processes. However, recent advancements in marine technology have introduced alternative ocean sampling techniques, such as subsurface floats, remotely operated vehicles (ROVs), and autonomous underwater vehicles (AUVs). These innovations have complemented traditional approaches, enabling more extensive and sustained ocean observations [1].

Among these emerging technologies, autonomous underwater gliders have become increasingly prominent due to their cost-effectiveness and operational efficiency. Gliders offer significant advantages over conventional methods, particularly for sustained, real-time oceanographic measurements that can span several months. These gliders autonomously move through the water column by altering their buoyancy, allowing them to change their weight relative to the surrounding water. This buoyancy adjustment generates an upward or downward force, resulting in a continuous cycle of gliding motion. Combined with changes in vehicle attitude, this leads to a sequence of ascent and descent, enabling the glider to traverse large ocean areas while minimizing energy consumption [1].

Known for their energy efficiency and long-duration operations in oceanographic applications [2], underwater gliders offer a scalable and cost effective solution for oceanographic observations. The concept of using fleets of autonomous underwater gliders for oceanographic research has advanced significantly over the years. A particularly promising application is the deployment of multiple gliders as part of a coordinated ocean sampling network to study specific oceanographic features. This idea has sparked extensive research into the automatic control of glider fleets, as noted by [3].

Leonard [3] presented a coordinated adaptive sampling method for the Adaptive Sampling and Prediction experiment, whose coordinate adaptive control capability of the Underwater Glider (UG) fleet was verified in Monterrey Bay, California [4]. Compared to a single underwater glider (UG), fleets of multiple vehicles demonstrate superior performance in ocean monitoring due to their enhanced coverage range, increased computing capacity, and improved data acquisition abilities. When applied to ocean observation tasks (OOT), UG fleets significantly enhance data accuracy and assimilation speed, enabling more comprehensive and reliable environmental monitoring [4]. The [5] paper reviews various communication technologies used in Underwater Sensor Networks (UWSNs), including acoustic, optical, and electromagnetic communication. Acoustic communication is most commonly used due to its long-range capabilities in underwater environments, while optical and electromagnetic methods are discussed in terms of their potential for high data rates and limited range. It highlights the challenges faced in underwater communication, such as signal attenuation, limited bandwidth, and the impact of environmental factors like water salinity and temperature. These challenges affect the reliability and efficiency of data transmission among AUVs. In [6] acoustic modems are currently the only technology capable of establishing long-range underwater communication, extending up to tens of kilometers. Despite their low bitrate due to the limited bandwidth of the acoustic channel, they remain the most commonly used in both industrial and research applications.

In [7] study on acoustic communication, the relationship between communication capacity and distance is profoundly influenced by the Signal-to-Noise Ratio (SNR). As distance increases, the SNR typically decreases due to the attenuation of the acoustic signal and the increase in background noise.The $\mathrm{SNR}\,(\mathrm{d}, \mathrm{f})$ as a function of distance $d$ and frequency $f$ in hertz is defined as,

$$\mathrm{SNR}\,(\mathrm{d}, \mathrm{f}) = \frac{P/A\,(d, f)}{N\,(f)\,\Delta\,(f)}$$

The received signal power P is measured in watts ($W$) or decibels relative to 1 $mW$ ($dBm$), representing the strength of the signal received at the receiver after it has propa-

gated through the medium, such as water in underwater communication. The propagation loss A (d,f), expressed in decibels ($dB$), quantifies the attenuation of the signal due to distance d and frequency f, accounting for losses incurred through absorption, scattering, and other degradation effects. The noise power spectral density N(f) is measured in watts per hertz (W/Hz) or decibels relative to 1 milliwatt per hertz ($dBm/Hz$) and represents the background noise present in the system that can interfere with signal reception. Lastly, the bandwidth $\Delta(f)$, indicated in hertz ($Hz$), refers to the range of frequencies occupied by the signal, affecting the amount of noise that can exist within that range. The SNR itself is a dimensionless quantity often expressed in decibels ($dB$) to provide a clearer understanding of the signal quality; a higher SNR indicates a stronger signal with less noise interference, which is crucial for effective communication, especially in challenging environments like underwater acoustics.

The reduction in SNR leads to a decrease in communication capacity, as the channel's ability to transmit data effectively is diminished. The thesis assumes that maintaining a high SNR is crucial for optimizing the performance of underwater acoustic communication systems. Effective communication over long distances requires advanced techniques to manage and enhance SNR, such as signal processing and noise reduction methods, to counteract the effects of signal degradation and ensure reliable data transmission.

The data transmission scheme in underwater acoustic communication involves a projector (transmitter) and a hydrophone (receiver). Initially, the projector collects sensor and navigational data, which it formats into packets at the data source. This data is then modulated with a carrier frequency to prepare it for transmission. The modulated signal is amplified to ensure it is strong enough to be effectively received. Once transmitted through the underwater environment, the signal reaches the hydrophone (receiver), which demodulates it to retrieve the original data. This process facilitates reliable data exchange in underwater communication systems by ensuring that information is accurately transmitted and received [8].

Once transmitted through the underwater environment, the signal reaches the hydrophone, which demodulates it to retrieve the original data. The effectiveness of this transmission is significantly influenced by the Signal-to-Noise Ratio (SNR). A high SNR is crucial for minimizing the impact of background noise and signal attenuation, thereby improving the clarity and reliability of the received data. As the signal travels through the water, factors such as noise and environmental conditions can degrade the SNR, affecting data accuracy. Thus, maintaining a high SNR is essential for ensuring effective and reliable underwater communication.

# CHAPTER 2
# Multi-Robot Sampling

## 2.1 Introduction

Multi-robot sampling involves coordinating multiple robots to efficiently explore and gather data from an environment. The primary goal is to enhance the coverage and quality of the collected data compared to using a single robot. This approach leverages the advantages of multiple robots, such as parallel exploration and redundancy, to achieve better performance in tasks like environmental mapping, search and rescue, or monitoring.

A key aspect of multi-robot sampling is the development of strategies for effective coordination among the robots. This involves designing algorithms that account for the robots' interactions, communication, and potential conflicts in their paths. Techniques such as decentralized control, communication protocols, and optimization methods are commonly used to manage these aspects.

For example, in the paper [9], the authors propose a method for coordinated multi-robot sampling in environmental monitoring applications. The approach uses a combination of sampling strategies and optimization algorithms to ensure that multiple robots cover the environment efficiently while minimizing overlap and maximizing data collection quality.

## 2.2 Background

In their comprehensive review [4], explore various technologies for underwater communication, highlighting the strengths and limitations of each. Optical communication utilizes light to transmit data underwater and offers high data rates with low latency. However, its effectiveness is constrained by the water's properties, such as absorption and scattering, which limit its range and reliability. Radio Frequency (RF) communication, while a potential option, suffers from significant attenuation in water, making it suitable only for very shallow depths and specialized contexts. In contrast, acoustic communication, which relies on sound waves, emerges as the most robust and widely adopted technology. Acoustic signals can travel long distances with moderate data rates, making them effective even in deep and turbid waters. Despite its slower speeds compared to optical methods, acoustic communication's ability to function reliably across a range of conditions solidifies its position as the preferred choice for most underwater communication systems. Compagnaro et al. underscore that while optical and RF technologies offer high performance in specific scenarios, acoustic communication remains the most viable option

for practical underwater applications due to its superior range and adaptability.

### 2.2.1 Modulation Techniques in Underwater Acoustic Communication

From [10]Efficient modulation techniques are essential in underwater acoustic communication (UWAC) systems to counteract the challenges of signal propagation, including noise and multipath effects. Three widely studied modulation methods—Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK), and 4-Quadrature Amplitude Modulation (4-QAM)—demonstrate unique advantages in various underwater scenarios.

### 2.2.2 Binary Phase Shift Keying (BPSK)

BPSK encodes data by shifting the phase of a carrier wave between two states (0° and 180°), representing binary values. Its signal can be represented as:

$$s(t) = A\cos(2\pi f_c t + \theta),$$

where $A$ is amplitude, $f_c$ is carrier frequency, and $\theta$ is the phase. BPSK offers robustness against noise, making it ideal for short-range, low-data-rate applications requiring reliable transmission with minimal complexity.

### 2.2.3 Quadrature Phase Shift Keying (QPSK)

QPSK improves upon BPSK by using four phase shifts (e.g., 0°, 90°, 180°, 270°) to encode two bits per symbol, doubling the data rate without increasing bandwidth. The signal is expressed as:

$$s(t) = A\cos(2\pi f_c t + \theta),$$

where $\theta$ is chosen from the four predefined angles. QPSK is well-suited for moderate data-rate applications where bandwidth efficiency is critical.

### 2.2.4 4-Quadrature Amplitude Modulation (4-QAM)

4-QAM combines amplitude and phase modulation to encode two bits per symbol. Its signal representation is:

$$s(t) = A_1\cos(2\pi f_c t + \phi_1) + A_2\sin(2\pi f_c t + \phi_2),$$

where $A_1, A_2$ are amplitudes and $\phi_1, \phi_2$ are phases. 4-QAM maximizes spectral efficiency, making it ideal for high-data-rate applications like video transmission in underwater environments.

### 2.2.5 Comparison and Applications

Each modulation technique caters to specific needs. BPSK excels in simplicity and noise resilience for low-data-rate applications. QPSK offers an effective balance between data rate and complexity, making it versatile for many underwater use cases. In contrast, 4-QAM maximizes throughput, suited for high-bandwidth requirements. Among these, QPSK often provides the best trade-off between spectral efficiency and robustness for general underwater communication.

# CHAPTER 3

# Impact of obstacles on line of sight (LOS) in Under- water Communications

In underwater environments, the phenomenon of multipath propagation is a significant challenge for acoustic communication. As sound waves travel from the transmitter to the receiver, they encounter various physical obstacles and environmental conditions that cause them to reflect, refract, and diffract along multiple paths. Each of these paths can have different attenuation factors depending on the specific medium and path characteristics. This means that the same signal, having traversed different paths, may arrive at the receiver at different times, leading to interference that can either amplify (constructive interference) or weaken (destructive interference) the received signal.

One of the key factors in underwater communication is the low propagation speed of sound waves in water, typically around 1500 meters per second. This is much slower than the speed of electromagnetic waves in air or vacuum, which causes several unique effects in underwater communication.

The authors in [11] omitted Doppler shift modeling in this paper due to the fixed transducer setup in open water without significant reflections. While this simplifies their model, it doesn't apply to scenarios with moving transmitters or environments with notable surface or bottom reflections.

The paper [11] presents a multipath diffraction model for ultrasonic pulse-echo systems used in underwater communication. The core objective is to account for how obstacles—both inside and outside the line-of-sight (LOS)—affect the propagation of acoustic signals. The model focuses on diffraction, reflection, and transmission phenomena and extends the knife-edge diffraction theory to acoustic environments. Where Line-of-sight(LOS) refers to the direct, unobstructed path between the transmitter (Tx) and receiver (Rx). When the obstacle is inside the LOS, it directly blocks or interferes with the signal path between the transmitter and receiver, whereas if the obstacle is outside the LOS, it doesn't interfere with the direct transmission path but may still affect the signal via diffraction or reflection.

**Fresnel-Kirchhoff Diffraction Parameter:** The model first introduces the Fresnel-Kirchhoff diffraction parameter v, which plays a crucial role in describing how obstacles affect signal propagation:

$$v = \frac{d_{los} * \sqrt{2\left(d_1 + d_2\right)}}{\lambda * 1 * d_1 * d_2}$$

$d_{los}$ is the line-of-sight distance between the transmitter TX and receiver RX. $d_1, d_2$ are the

distances from the TX to the obstacle and from the obstacle to RX. $\lambda$ is the wavelength of the ultrasonic signal calculated as $\lambda = \frac{c}{f}$ where c is the speed of sound in water ( 1500 m/s), and f is the signal frequency. This parameter helps define the diffraction

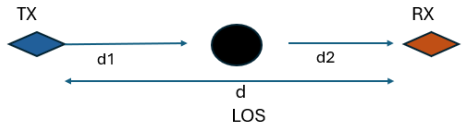


**Figure 3.1.** Transmission of signal in presence of obstacle with direct Line of Sight (LOS)

characteristics as the signal interacts with the obstacle.

**Diffraction from the Obstacle**: The paper calculates the diffraction effects by considering the obstacle's shape (in this case, a cylindrical obstacle) and the pressure intensity at both the front and back of the obstacle. The pressure intensity at the front and back is defined by:

$F_{front}\left(v_{front}\right)$and $F_{back}\left(v_{back}\right)$

These components contribute to the overall attenuation and phase shift caused by the obstacle. Transmission Factor for Obstacles Inside and Outside LOS: For obstacles inside the LOS, the cross-section of the obstacle that intersects the line-of-sight is calculated as:

$$d_{obstacle}\left(d_{los}\right) = 2 * cos\left(\frac{d_{los}}{\frac{b}{2\pi}}\right)$$

Where b is the diameter of the obstacle and $d_{los}$ is the distance along the line-of-sight.

## 3.1 Modelling Attenuation with Obstacle

In this section, we model the attenuation of an underwater ultrasonic signal in the presence of a cylindrical obstacle. The model incorporates diffraction, line-of-sight (LOS) transmission, and reflection components, as described in the referenced paper. The attenuation is calculated as a function of frequency and obstacle position.

### 3.1.1 Diffraction

Diffraction effects are modeled using the Fresnel integral, computed separately for the front and back of the obstacle:

$$F_{\text{front}}\left(t, d_{\text{LOS}}\right) = F\left(v_{\text{front}}\right) \cdot \sin\left(2\pi f t + \arctan\left(\frac{\operatorname{Im}(F)}{\operatorname{Re}(F)}\right)\right),$$

$$F_{\text{back}}\left(t, d_{\text{LOS}}\right) = F\left(v_{\text{back}}\right) \cdot \sin\left(2\pi f t + \arctan\left(\frac{\operatorname{Im}(F)}{\operatorname{Re}(F)}\right)\right),$$

where $v_{\text{front}}$ and $v_{\text{back}}$ are the Fresnel-Kirchhoff parameters for the front and back, respectively:

$$v_{\text{front}} = \frac{d_{\text{LOS,front}} \cdot \sqrt{2\left(d_1 + d_2\right)}}{\lambda \cdot d_1 \cdot d_2}, \quad d_{\text{LOS,front}} = d_{\text{LOS}} - \frac{b}{2},$$
$$v_{\text{back}} = \frac{d_{\text{LOS,back}} \cdot \sqrt{2\left(d_1 + d_2\right)}}{\lambda \cdot d_1 \cdot d_2}, \quad d_{\text{LOS,back}} = d_{\text{LOS}} + \frac{b}{2}.$$

The complex Fresnel integral is defined as:

$$F\left(v\right) = 1 + j\left(\left(0.5 - C\left(v\right)\right) - j\left(0.5 - S\left(v\right)\right)\right),$$

where $C\left(v\right)$ and $S\left(v\right)$ are the Fresnel cosine and sine integrals.

### 3.1.2 LOS Transmission

The LOS transmission factor models the signal passing directly through the obstacle:

$$F_{\text{LOS}}\left(t, d_{\text{LOS}}\right) = \begin{cases} F_0 \cdot \cos\left(\frac{\pi d_{\text{LOS}}}{b/2}\right), & \left|d_{\text{LOS}}\right| \leq b/2, \\ 0, & \text{otherwise.} \end{cases}$$

### 3.1.3 Reflections

Reflection components account for signal paths bouncing off nearby surfaces:

$$F_{\text{refl}}\left(t\right) = R_1 \cdot \sin\left(2\pi f t + \pi\right) \cdot w_{\text{Hann}}\left(t - \tau_{\text{refl}}\right),$$

where $R_1$ is the reflection factor, $\tau_{\text{refl}} = \frac{d_{\text{refl}}}{c}$ is the propagation delay, and $w_{\text{Hann}}\left(t\right)$ is the Hanning window:

$$w_{\text{Hann}}\left(t\right) = 0.5 \cdot \left(1 - \cos\left(\frac{2\pi t}{T_{\text{pulse}}}\right)\right).$$

### 3.1.4 Total Attenuation

The total attenuation is the superposition of all components:

$$F_{\text{total}}\left(t, d_{\text{LOS}}\right) = F_{\text{front}}\left(t - \tau_{\text{front}}, d_{\text{LOS}}\right) + F_{\text{back}}\left(t - \tau_{\text{back}}, d_{\text{LOS}}\right) + F_{\text{LOS}}\left(t - \tau_{\text{LOS}}, d_{\text{LOS}}\right) + F_{\text{refl}}\left(t - \tau_{\text{refl}}\right).$$

The magnitude of $F_{\text{total}}$ is converted to decibels (dB) as,

$$\text{Attenuation (dB)} = 20 \cdot \log_{10}\left(\left|F_{\text{total}}\right|\right).$$

# CHAPTER 4
# Simulation

The simulation is performed over a frequency range of 100–135 kHz and obstacle positions $d_{\text{LOS}}$ between 0 and 12 cm. The parameters are:

- $c = 1503\,\text{m/s}$ (speed of sound in water),
- $b = 0.006\,\text{m}$ (obstacle diameter),
- $d_1 = 0.062\,\text{m}$, $d_2 = 0.062\,\text{m}$ (fixed distances),
- $R_1 = 0.15$ (reflection factor),
- $T_{\text{pulse}} = 92.71\,\mu\text{s}$ (pulse duration).

The attenuation results are visualized as a heatmap with:

- X-axis: Frequency (kHz),
- Y-axis: Obstacle position $d_{\text{LOS}}$ (cm),
- Color: Attenuation (dB).

### 4.0.1 Effect of Removing Front and Back Terms in Attenuation Model

If the front and back diffraction terms are removed from the attenuation model, the resulting attenuation plot would be significantly simplified. Without these terms, the model no longer accounts for the diffraction effects caused by the obstacle's edges, leading to the following changes:

- **Simplified Attenuation Profile:** The plot would primarily show attenuation as a function of obstacle position ($d_{\text{LOS}}$) and frequency, driven by the line-of-sight (LOS) transmission and reflection components. The alternating horizontal bands resulting from constructive and destructive interference due to edge diffraction will no longer appear.
- **LOS-Driven Attenuation:** Attenuation will depend on whether the obstacle intersects the LOS. For obstacle positions within the LOS, attenuation will vary smoothly based on the fraction of the LOS blocked by the obstacle. Beyond the LOS, attenuation will predominantly depend on the reflection component.
- **Reflection Contribution:** The reflection term will still contribute periodic attenuation effects based on the frequency and the geometry of the reflection paths. However, the absence of edge diffraction effects will reduce the complexity of the attenuation profile.

Overall, the plot would depict a smoother attenuation pattern dominated by LOS transmission and reflection. While this simplifies the model and its visualization, it excludes the intricate effects of edge diffraction, which are critical for accurately modeling

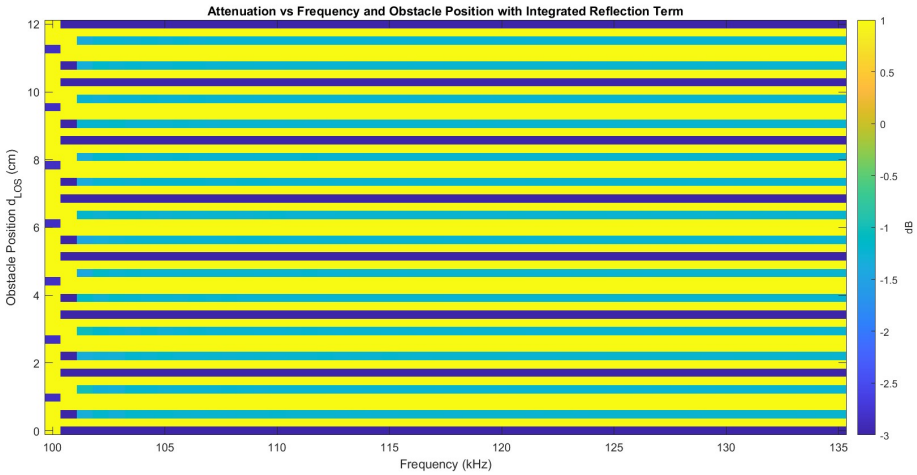


**Figure 4.1.** Heatmap of Attenuation (dB) as a Function of Frequency and Obstacle Position

the interaction of ultrasonic waves with obstacles.

### 4.0.2 Adaptive sampling:

Autonomous underwater vehicles (AUVs) and autonomous surface vehicles (ASVs) have been employed for area coverage tasks, often using predetermined paths like lawnmower patterns (Galceran Carreras, 2013). These methods are inefficient for detecting and localizing regions of interest (ROIs) over large areas, especially in marine environments where communication limitations hinder real-time trajectory updates. Consequently, data collected by these systems is typically analyzed post-mission. However, recent advancements enable autonomous systems to adapt their trajectories in real-time based on in situ data processing, allowing for focused sampling on high-level objectives like ROI localization. Researchers have developed adaptive sampling strategies that leverage the spatial correlation of ROIs to enhance robotic environmental monitoring (Low et al., 2009; Mysorewala et al., 2009). These strategies use sensor data collected during the vehicle's journey to update an environmental model, which then guides subsequent sampling paths.

In [12] building on a previous system by Tan et al. (2018), which was limited to simpler 2D environments for surface vehicles, this study extends the methodology to 3D environments. The authors introduce a Gaussian Process (GP)-based adaptive sampling method designed to balance exploration and exploitation by dynamically adjusting sampling density based on sensor data. These strategies incorporate sensor information collected during the vehicle's trajectory into an environmental model that guides future sampling paths, thus enhancing the efficiency and effectiveness of data collection. The study introduces two informative path planning methods—branch and bound techniques and cross-entropy optimization—to select future sampling locations in a three-dimensional environment while considering the motion constraints of the system. These methods were tested on an underactuated AUV through simulations and preliminary field experiments in the Severn River, MD, demonstrating their capability to prioritize ROI measurements

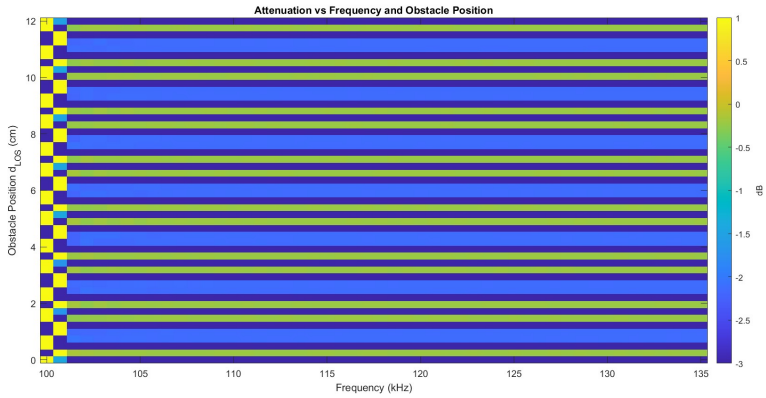


**Figure 4.2.** Attenuation as a function of frequency and obstacle position ($d_{\mathrm{LOS}}$) considering only LOS transmission and reflection components, excluding diffraction effects from the obstacle's front and back edges.

without compromising the overall model fidelity of the full sampling area. The authors identify several promising directions for future research. These include improving planning efficiency by developing intelligent surfacing strategies, reducing travel distance and mission time, incorporating tidal current knowledge, and extending the framework to multi-vehicle teams. Such enhancements could further increase the efficiency and robustness of adaptive sampling in diverse marine environments.

### 4.0.3 Model Predictive Control

Model Predictive Control (MPC) is a control strategy that optimizes control actions by solving a finite-horizon optimization problem at each control step. The key idea behind MPC is to use a model of the system to predict future states and control inputs over a defined prediction horizon. At each time step, MPC solves an optimization problem to minimize a cost function, which typically includes terms for tracking errors and control effort. The optimal control inputs are applied to the system, and the process is repeated at the next time step with updated measurements.MPC offers several advantages, including the ability to handle multi-variable systems with constraints on inputs and states. It is widely used in various applications, such as robotics, automotive systems, and industrial processes, due to its ability to handle complex constraints and provide robust performance under varying conditions [13].

## 4.1 Adaptive Control for High SNR Communication

### 4.1.1 Problem Statement:

This project aims to develop an optimal control strategy for underwater gliders engaged in a cooperative sampling of regions of interest(ROIs) within a two-dimensional space

populated with obstacles. The gliders must determine and move toward the most suitable ROI while maintaining reliable communication quantified in terms of a minimum SNR.

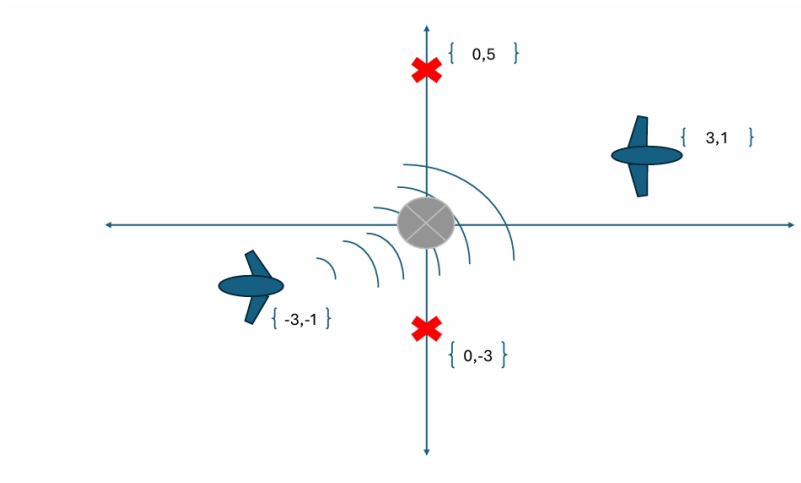


**Figure 4.3.** Two underwater gliders in an environment with an obstacle.

The decision-making process involves evaluating the performance of each glider based on three key factors:distance to ROI, effort required and communication quality. To ensure effective and reliable data exchange, the communication quality is assessed using the signal-to-noise ratio(SNR), which serves as a critical component in the Gradient descent framework. Higher SNR will indicate better quality, essential for coordinated operations.

For effective data exchange, communication quality is determined by the SNR, with higher values indicating better quality. The gliders must remain within communication range to ensure continuous data exchange and coordinated operations. The system must account for signal attenuation, avoid high-attenuation areas, and consider multipath effects caused by obstacles, which can degrade signal quality. These considerations are crucial for planning paths and selecting ROIs to maintain reliable communication.

## 4.2 Vehicle Dynamics:

The glider operates exclusively in a 2-dimensional plane, with thrusters providing linear force output proportional to PWM input commands. Lateral/yaw motion and roll dynamics are neglected, simplifying control to pitch and depth adjustments. The glider's IMU sensor delivers real-time position and orientation data (pitch, roll, yaw), enabling closed-loop navigation without external localization aids. Thruster responses are assumed instantaneous and frictionless, while hydrodynamics follow quasi-static models (drag/lift linear with velocity). Mass redistribution via stepper motors occurs without delay, enabling immediate center-of-gravity shifts for pitch control. Environmental factors like water density variations and turbulence are excluded, and communication between subsystems (Pi-Pixhawk-Arduino) is assumed lossless.

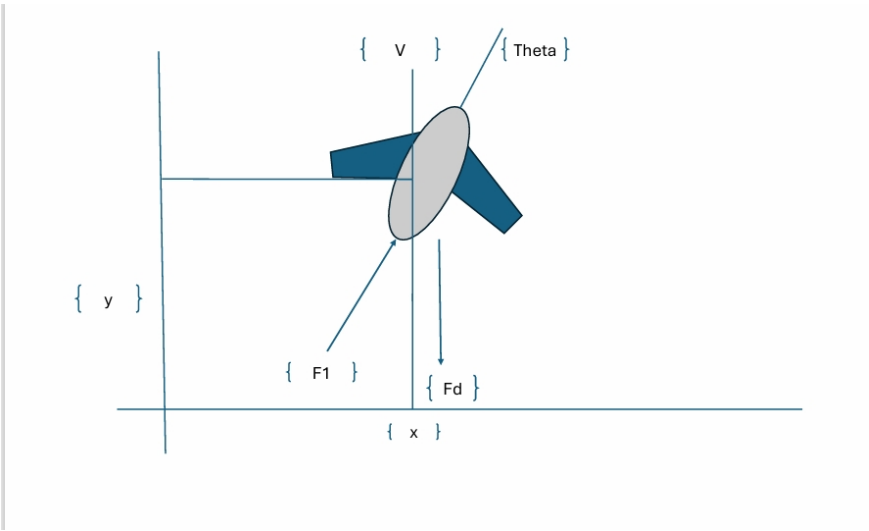


**Figure 4.4.** Glider

Assuming that the thrusters provides force along the directions they are mounted, the system represents a unicycle model, which is a simplified model often used for robotic vehicles or gliders. The system has three states: (a) $x$, defining $x$- position of glider, (b) $y$, defining $y$- position of glider and, (c) $\theta$ angular orientation of the glider in $x-y$ plane. The control inputs are $u$, forward velocity of the glider and, $\omega$ angular velocity of the glider.

The system dynamics can then be defined as,

$$[\dot{x} = u \cdot \cos(\theta)]$$

$$[\dot{y} = u \cdot \sin(\theta)]$$

$$[\dot{\theta} = \omega]$$

Including SNR, how SNR changes over time as the glider moves.
For simplicity:
$SNR = \frac{P_t}{N_0 d^2}$ from [14] The Equation depicts a simple relationship between SNR and desitance, where the SNR decreases with increasing distnaces. Here k is a constant that could represent the degradation of SNR overtime due to multipath.
Cost function including SNR:

$$J = \sum_{k=0}^{N-1} [{x_k}^T Q x_k + {u_k}^T R u_k + q_{SNR} (SNR_{des} - SNR)^2 \, dt]$$

## 4.3 Optimal Control for Multi-AUV System

### 4.3.1 Gradient Descent

Considering that we have two underwater gliders with the same dynamics for both of them, the cost function will now include the state, control, and communication costs for

both glides.

$$J = \sum_{k=0}^{N-1} [{x_{1k}}^T Q x_{1k} + {x_{2k}}^T Q x_{2k}] + [{u_{1k}}^T R u_{1k} + {u_{2k}}^T R u_{2k}] + q_{SNR} \left(SNR_{des} - SNR\right)^2 dt]$$

Firstly just the gliders moving toward their desired position positions control law using the gradient descent method.

$$J = \sum_{k=0}^{N-1} \left[ \left(x_{k1} - x_{k1}^{des}\right)^T Q \left(x_{k1} - x_{k1}^{des}\right) + \left(x_{k2} - x_{k2}^{des}\right)^T Q \left(x_{k2} - x_{k2}^{des}\right) \right] \\ + dt \left[ \left(u_{k1}^T R u_{k1} + u_{k2}^T R u_{k2}\right) + q_{SNR} \left(SNR_{des} - SNR\right)^2 \right]$$

Here's what each term represents: - $\left(x_{k1} - x_{k1}^{des}\right)^T Q_p \left(x_{k1} - x_{k1}^{des}\right)$ penalizes the deviation of the first glider's position from its desired position. - $\left(x_{k2} - x_{k2}^{des}\right)^T Q_p \left(x_{k2} - x_{k2}^{des}\right)$ penalizes the deviation of the second glider's position from its desired position. - $u_{k1}^T R u_{k1}$ and $u_{k2}^T R u_{k2}$ represent the cost associated with the control inputs for the first and second gliders, respectively. - $q_{SNR} \left(SNR_{des} - SNR\right)^2$ represents the cost associated with the deviation of the actual SNR from the desired SNR. This cost function now includes terms to penalize deviations from desired positions, in addition to the costs related to control inputs and SNR deviations.

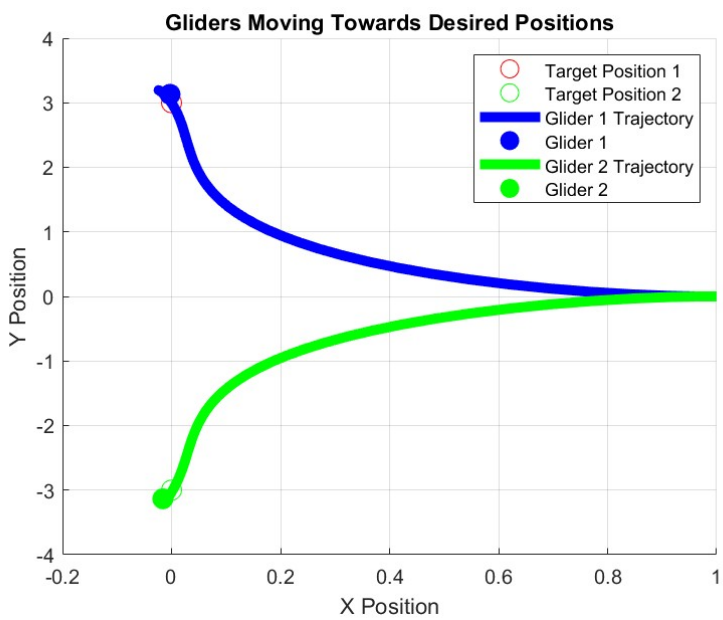


**Figure 4.5.** Same initial point Gradient Descent.

### 4.3.2 Model Predictive Control

**System Setup and Dynamics:** The system represents a unicycle model, which is a simplified model often used for robotic vehicles or gliders.

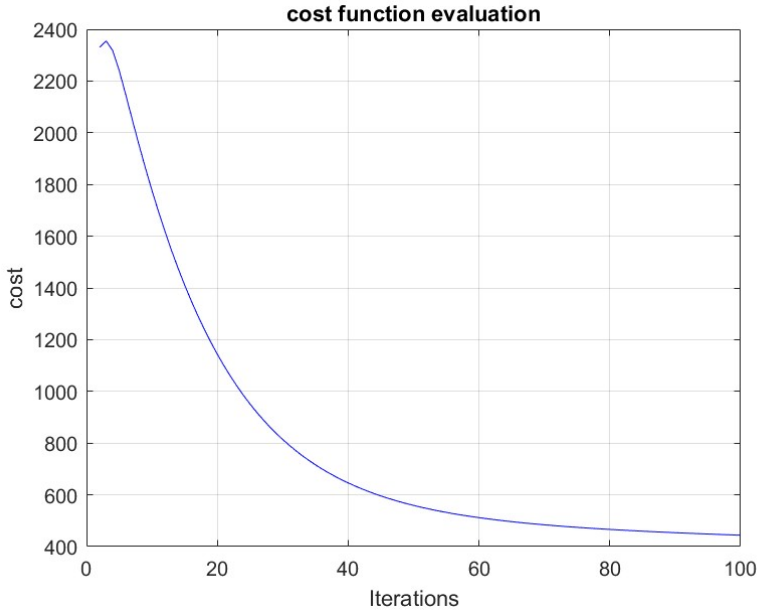


**Figure 4.6.** Cost Evaluation with same initial points Gradient Descent

**Model Predictive Control (MPC) setup:** Prediction Horizon (N): This is the number of time steps into the future that the MPC controller considers when making decisions. In the code, (N = 10).Cost Function:The MPC controller aims to minimize a cost function that balances two objectives:Minimizing the distance between the glider's current position and a desired position.Minimizing the control effort (i.e., the values of $u$ and $\omega$.The cost function is defined as:

$$mterm = [(x - x_{\text{desired}})^2 + (y - y_{\text{desired}})^2]$$

$$lterm = [(x - x_{\text{desired}})^2 + (y - y_{\text{desired}})^2 + R \cdot \left(u^2 + \omega^2\right)]$$

here, $Q$ and $R$ are weighting matrices that balance the importance of state errors and control effort.Constraints:The control inputs are constrained to ensure the glider operates within realistic bounds:(u) is constrained between 0.0 and 1.0 , the glider can only move forward.$\omega$ is constrained between $-\frac{\pi}{4}$ and $\frac{\pi}{4}$, limiting the maximum turn rate.

**MPC Execution and Simulation:**

- **Initialization:** The initial state of the glider is set to [0, 0, 0], meaning it starts at the origin facing along the x-axis.
- **MPC Loop:** The MPC controller computes the optimal control inputs (u) and $\omega$ by solving the optimization problem over the prediction horizon.The system dynamics are updated using the computed control inputs.The glider's position and orientation are stored for later analysis.Plotting Results:After

the simulation, the trajectory of the glider is plotted, showing how it moves towards the desired position over time.

# CHAPTER 5

# Hardware

## 5.1 Power consumption analysis and runtime limitation

This section provides a detailed analysis of the electronic components and their power requirements for an underwater glider. A significant finding is the high total current draw of $41.87A$ when all components operate simultaneously, primarily driven by the two T200 thrusters ($34A$ combined). With a $14.8V$, $10{,}000mAh$ ($10Ah$) lithium battery, the estimated runtime is only 0.24 hours (14.4 minutes). This severe limitation highlights the critical need for power optimization strategies, such as duty cycling thrusters or selecting lower-power alternatives, to extend mission duration.

### 5.1.1 Component selection and specifications

Specific components required are listed here.

- **Processing Units:**
  - Raspberry Pi 2 (Cortex-A7, 1GB RAM) handles complex navigation but draws $0.8A$ and lacks real-time guarantees.
- **Actuation System:**
  - Propulsion: Two Blue Robotics T200 thrusters with ESCs, requiring 12-16$V$ for optimal thrust-to-power ratio.
  - Pitch/Roll Control: OSUG-inspired mechanism using two NEMA 17 steppers:
    - Pitch: $38mm$ motor ($1.5A$) moves a $713g$ battery mass axially via lead screw.
    - Roll: $60mm$ motor ($1.8A$) rotates pitch assembly via planetary gearbox.
- **Sensors:**
  - Bar30 Pressure Sensor: 2.5–5.5$V$ supply, $3.3V$ logic, requires $4.7k\Omega$ I²C pull-ups.
  - Temperature Sensor: 3.3–5.5V, 3.3V logic-only.
  - Leak Sensor: 3.3–5V operation.
  - Low-Light USB Camera: 5V, 220mA max.

### 5.1.2 System Architecture

The glider employs a distributed architecture.

- Raspberry Pi 4 acts as the central hub, processing sensor data, running navigation algorithms, and coordinating subsystems. It communicates via I2C with sensors and the Arduino, and MAVLink/UART with the Pixhawk.
- Pixhawk Flight Controller manages thruster control through PWM signals to ESCs, powered directly by the main battery.
- Arduino Uno with a CNC Shield drives the NEMA 17 stepper motors via A4988 drivers, executing pitch/roll commands from the Pi.

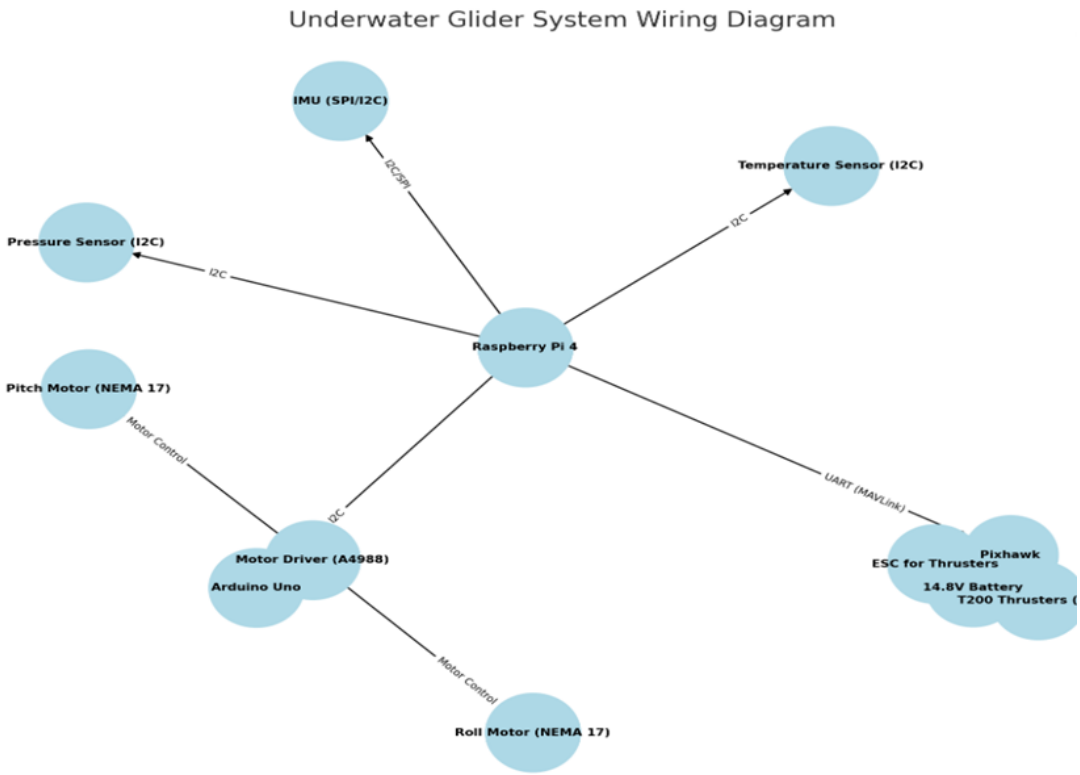


**Figure 5.1.** Control Architecture

## 5.2 Implementation challenges and solutions

Initial breadboard testing revealed excessive motor vibrations, resolved by implementing micro-stepping and transitioning to a CNC Shield for stable motor control. Sensor integration required careful voltage matching: logic-level converters were initially used for 3.3V sensors (e.g., Bar30) with the 5V Arduino, but became unnecessary when the Pi (native 3.3V I2C) took over sensor management. Thruster calibration was performed via Pixhawk's PWM outputs.

The pitch/roll mechanisms draw inspiration from OSUG (open source underwater glider) design, using movable masses and geared sub-assemblies for buoyancy adjustment and directional control. Testing involves SSH access to the Pi, UART/I²C communication validation, and custom Python scripts (using pymavlink and Blue Robotics libraries) for sensor data acquisition and system monitoring.

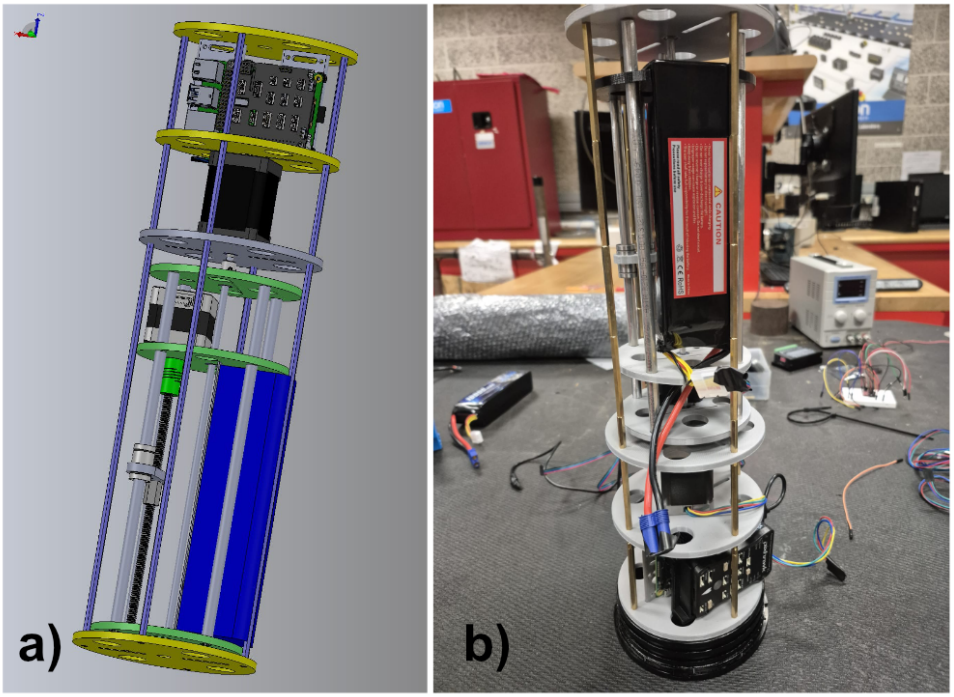


**Figure 5.2.** Images of a) CAD model of the internals of the glider and, b) assembled internal parts of the glider.

# CHAPTER 6
# Timeline

### 6.0.1 Fall 2024

**Objective** Develop foundational models for underwater acoustic communication, test modulation techniques, and initiate electronics and PCB design for the AUV.

1. **Formulate a Detailed SNR Model:**
   - Develop a mathematical model for SNR considering distance, obstacles, and frequency attenuation.
   - Test modulation techniques such as Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK).
   - Simulate the performance of modulation techniques using MATLAB or Python.
   - *Deliverable:* Validated SNR model with comparative analysis of modulation schemes.
2. **Simulate AUV Dynamics with Basic 2D Models:**
   - Create a simulation environment for AUVs with basic 2D motion (linear and angular).
   - Test communication models under open water and obstacle-heavy scenarios.
   - *Deliverable:* Simulation results highlighting the impact of modulation schemes in different environments.
3. **Extend the Model to Multiple AUVs:**(Holidays)
   - Scale the simulation to include 3–5 AUVs interacting in 2D.
   - Study the effects of interference and scalability on communication performance.
   - *Deliverable:* Report summarizing the performance of multiple AUVs using different modulation techniques.
4. **Electronics and PCB Design Initiation:**
   - Design initial schematics for motor drivers, sensor connections, and communication systems using Eagle or Altium Designer.
   - Integrate acoustic transducers, pressure sensors, leak sensors, and power systems.
   - *Deliverable:* Preliminary PCB design schematics for subsystem testing.

### 6.0.2 Spring 2025

**Objective:** Introduce realistic 3D dynamics and localization in simulation, finalize PCB design, and prepare for pool experiments.

1. **Introduce Realistic Dynamics in 3D Simulation:**
   - Upgrade simulation to include vertical motion (depth), pitch, roll, and yaw.
   - Test control algorithms such as PID and Model Predictive Control (MPC) for trajectory optimization.
   - *Deliverable:* Validated 3D simulation incorporating realistic dynamics and communication models.
2. **Finalize Electronics and PCB Design:**
   - Complete PCB designs for motor drivers, communication interfaces, and power regulation circuits.
   - Test the functionality of integrated sensors and subsystems.
   - *Deliverable:* Fully assembled and tested electronics system with functional PCBs.
3. **Prepare Experimental Plan:**
   - Design protocols for pool experiments, including visual light signaling for communication.
   - Set up and calibrate a stereo GoPro camera system for real-time position and orientation tracking.
   - *Deliverable:* Comprehensive experimental protocol for pool testing.

### 6.0.3 Summer 2025

**Objective:** Conduct comprehensive simulations and real-world pool experiments to validate the system.

1. **Comprehensive Simulation Tests:**
   - Test the impact of obstacle density and communication scalability on performance.
   - Evaluate the efficiency of various modulation techniques under different environmental conditions.
   - *Deliverable:* Simulation results summarizing modulation performance and system scalability.
2. **Pool Experiments:**
   - Test real-world communication and navigation performance using the designed AUVs.
   - Validate control algorithms and localization techniques using the stereo camera system.

- Test the durability and functionality of electronics and PCBs in a controlled underwater environment.
- *Deliverable:* Detailed experimental report comparing simulation and real-world results, with recommendations for improvements.

### 6.0.4 Updated project summary

- Number of gliders: Originally, the plan involved single autonomous glider, to effectively demonstrate cooperative behavior and to meet the scope of the thesis to show the control of multiple autonomous gliders and real-time control.
- Control strategy: The initial plan was to implement model predictive control as the primary control approach. Now, the project focuses on Distributed Coverage control as the main strategy for area exploration and sampling. However, MPC will still be used, but specially only for local obstacle avoidance, complementing the global distributed strategy.
- Testing Environment: Testing was initially planned in the Anderson pool on campus. Due to its closure, testing will be conducted in a nearby lake. However, because the lake is murky, the communication and the video quality for the glider will have constraints.
- Communication Method: Acoustic modems were initially considered for underwater communication. Due to budget constraints, the project shifted towards using RF (Radio Frequency) communication.
- Modulation techniques: Instead of relying on a single method, the project will now evaluate different modulation techniques.
- Scalability: After validating the setup with two gliders, the simulation and methodology will be extended to multiple gliders, enabling broader analysis and distributed control performance in larger networks.

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